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PUBLIC KEY ENCRYPTION

Public key encryption isalso called Asymmetric encryption. It uses the concept of a key value pair, a different key is used for the encryption and decryption process. One of the keys is typically known as the private key and the other is known as the public key.

The private key is kept secret by the owner and the public key is either shared amongst authorized recipients or made available to the public at large.

The data encrypted with the recipient's public key can only be decrypted with the corresponding private key. **Figure 17.11** shows the public key encryption.

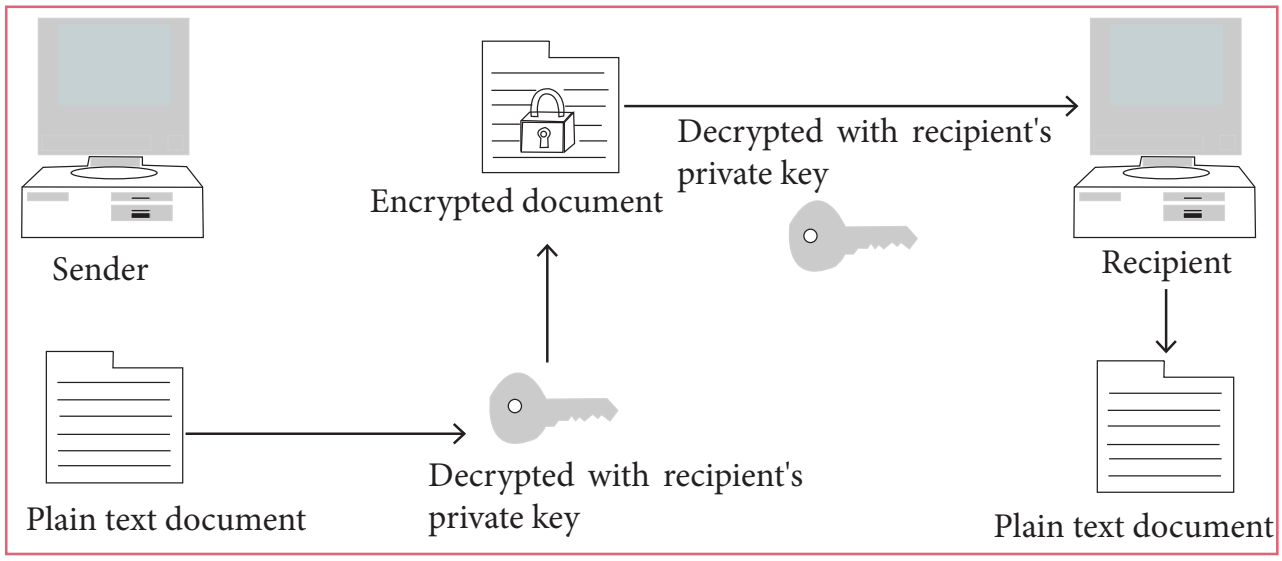

Figure 17.11 Public key encryption

Asymmetric Encryption in Digital Certificates:

A digital certificate in a client-server model of communication is one of the example of Asymmetric Encryption. A certificate is a package of information that identifies a user and a server. It contains information such as an organization's name, the organization that issued the certificate, the users' email address and country, and user's public key.

When a server and a client require a secure encrypted communication, they send a query over the network to the other party, which sends back a copy of the certificate. The other party's public key can be extracted from the certificate. A certificate can also be used to uniquely identify the holder.

Digital Signature

Digital signatures are based on asymmetric cryptography and can provide assurances of evidence to origin, identity and status of an electronic document, transaction or message, as well as acknowledging informed by the signer.